# Empirical Project - Final Report

Do Quantitative Easing and Quantitative Tightening have asymmetric effects on bond and equity risk premia in Europe?

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**Abstract**

This paper examines whether Quantitative Easing (QE) and Quantitative Tightening (QT) have asymmetric effects on bond and equity risk premia in the euro area. Using the high-frequency shock identification of Altavilla et al. (2019) combined with panel local projections (Jordà, 2005) over September 2004 to September 2025, the ECB's balance-sheet surprise is interacted with mutually exclusive regime dummies to estimate dynamic effects on sovereign spreads and sectoral equity risk premia. Three main results emerge. First, QT surprises generate persistent sovereign spread widening extending to the one-year horizon, concentrated in peripheral economies, while QE compression is transitory and dissipates within one quarter. Second, QT shocks produce a more durable depression of equity risk premia than equivalent QE shocks, with the negative effect remaining statistically significant at twelve months under QT but fully reverting under QE. Third, no equity-duration heterogeneity is detected across sectors, pointing to a risk-taking channel rather than differential term-premium compression. These findings suggest that QT tightens financial conditions more durably than QE eases them, with direct implications for the pace and communication of the ECB's balance-sheet normalisation.

**Keywords:** Quantitative Easing, Quantitative Tightening, Risk Premia, Panel Local Projections, Euro Area

# 1 Introduction

The past two decades have witnessed an unprecedented expansion of central bank balance sheets. Following the Global Financial Crisis of 2008, the European Central Bank (ECB) embarked on a series of large-scale asset purchase programmes, from the Covered Bond Purchase Programme to the Asset Purchase Programme and the Pandemic Emergency Purchase Programme, that collectively brought its balance sheet to nearly €9 trillion at its peak in 2022. The rationale was well understood: with policy rates constrained by the effective lower bound, large-scale asset purchases, commonly referred to as Quantitative Easing (QE), were expected to compress long-term yields, reduce sovereign risk premia, and stimulate risk-taking across financial markets through the portfolio-balance and signalling channels. Since June 2022, however, the ECB has been navigating the challenge of Quantitative Tightening (QT): the gradual reduction of its securities holdings in the face of persistent inflation. This reversal raises a question of both academic and immediate policy relevance: *are the financial market effects of QT simply the mirror image of QE, or does monetary policy transmission operate asymmetrically across expansion and contraction phases?*

This question is consequential for at least two reasons. First, if QT surprises generate more persistent financial tightening than equivalent QE surprises generate easing, then the effective tightening per unit of balance-sheet reduction may be larger than historical QE episodes would suggest, with direct implications for the pace and communication of the ECB's normalisation path. Second, the euro area context introduces a dimension absent from the US Treasury market studied by Lloyd and Ostry (2024): sovereign fragmentation risk. Asset purchases have historically served as an implicit backstop for peripheral sovereign debt, compressing spreads between, say, Italian BTPs and German Bunds. Whether the withdrawal of this backstop through QT generates a symmetric decompression, or a disproportionate and persistent widening in vulnerable economies, is a first-order concern for the Eurosystem's financial stability mandate.

To address these questions, we combine the high-frequency shock identification of Altavilla et al. (2019) with panel local projections (Jordà, 2005), pooled across countries and sectors over a sample spanning September 2004 to September 2025. We interact the ECB's balance-sheet surprise with mutually exclusive regime dummies to isolate QE and QT shocks, and estimate their dynamic effects on two families of outcome variables: 10-year sovereign spreads relative to the German Bund, and sectoral equity risk premia for Banks, Technology, Utilities, and Oil & Gas.

Our analysis delivers four main findings. First, QT surprises generate a persistent sovereign spread widening that extends to the one-year horizon and is concentrated in peripheral economies, while the compressive effect of QE surprises is transitory and dissipates within one quarter. Second, QT shocks produce a more durable depression of equity risk premia than equivalent QE shocks: the negative impact on cumulative excess returns remains statistically significant at twelve months under QT, whereas it fully reverts under QE. Third, we find no evidence of equity-duration heterogeneity: QE and QT shocks affect all sectors uniformly, pointing to a risk-taking channel rather than differential term-premium compression along the cash-flow maturity structure. Fourth, a sensitivity analysis restricting the QE regime to the formal Asset Purchase Programme reveals a dual asymmetry: APP surprises generate substantially larger immediate effects on both spreads and equity risk premia than QT surprises of equal magnitude, yet these effects dissipate within two to three quarters, whereas QT effects persist to the one-year horizon: a magnitude asymmetry at short horizons coexisting with a persistence asymmetry at long horizons.

# 2 Literature Review

This paper draws on three strands of literature: the identification of monetary policy surprises, the asymmetric effects of QE and QT on financial markets, and the local-projection method for estimating dynamic causal effects.

## 2.1 High-Frequency Identification of Monetary Policy

The identification of monetary policy shocks from high-frequency asset-price movements around central bank announcements has become a standard approach in the literature. Altavilla et al. (2019) apply this framework to the euro area and construct the Euro Area Monetary Policy Event-Study Database (EA-MPD). Exploiting the institutional separation between the ECB's press release at 13:45 CET and the press conference at 14:30 CET, they use factor analysis on intraday OIS rate changes to identify four dimensions of ECB communication: a target factor for the current policy rate, a timing factor for near-term rate expectations, a forward guidance factor for the medium-term path, and a QE factor loading on the long end of the yield curve. Their dataset has become the standard tool for measuring euro-area monetary policy surprises. Swanson (2021) develops an analogous decomposition for the United States, extracting three orthogonal factors from asset-price movements in 30-minute windows around FOMC announcements: a federal funds rate factor, a forward guidance factor, and an LSAP factor capturing balance-sheet policy surprises. He finds that forward guidance and LSAPs had effects on financial markets comparable in magnitude to conventional rate changes.

## 2.2 Asymmetric Effects of QE and QT

Lloyd and Ostry (2024) directly test whether QE and QT have symmetric effects on US Treasury yields. Using the LSAP surprises of Swanson (2021), they estimate daily-frequency local projections and interact the shocks with indicators for QE, full-reinvestment, and asset-runoff periods. Their central finding is that QT surprises during the asset-runoff phase (2017 to 2019) had larger and more persistent effects on 2-year yields than QE surprises of equal magnitude. Using three alternative decompositions of yields into rate expectations and term premia, they show that the asymmetry arises from the expectations component, suggesting that the signalling channel operates more powerfully during tightening. They attribute this to the effective lower bound, which constrains downward revisions of rate expectations during QE but imposes no such ceiling during QT. Our paper extends their analysis to the euro area, where the additional dimension of sovereign fragmentation risk provides a distinct channel for asymmetric transmission.

### 2.3 Local Projections

Jordà (2005) introduces the local-projection method as a flexible alternative to VARs for estimating impulse response functions. The approach estimates a separate regression at each forecast horizon, which avoids propagating misspecification errors across horizons and naturally accommodates non-linear specifications such as regime-dependent or interaction effects. These properties make it well suited to our setting, where we test for asymmetry between QE and QT shocks and for heterogeneity across countries and sectors. The combination of high-frequency monetary policy surprises with local projections at lower frequency, following the proxy-SVAR logic of Gertler and Karadi (2015), has become a standard approach in the empirical monetary policy literature.

### 2.4 Contribution

Our paper brings the QE-versus-QT asymmetry question to the euro area, combining the high-frequency identification of Altavilla et al. (2019) with panel local projections pooled across countries and sectors. While Lloyd and Ostry (2024) focus exclusively on US Treasury yields, we jointly examine sovereign bond spreads and sectoral equity risk premia, and we test whether the portfolio-balance channel generates heterogeneous effects along the equity-duration spectrum.

## 3 Data Description

The raw database contains 673 observations for 53 variables. Each observation corresponds to a specific month. After data cleaning and creating some variables, the database consists of 253 observations for the same 86 variables. The time span is from September 2004 to September 2025. The different variables in this database are grouped by category and presented below, along with their sources. A codebook is also available in the Appendix to describe all these variables.

### 3.1 Data Presentation

**Bond market data:** The dataset includes a comprehensive set of sovereign and risk-free bond market indicators for the euro area. Long-term sovereign borrowing costs are measured using 10-year government bond yields for Germany, France, Italy, Spain, and Portugal. In addition, the euro area zero-coupon yield curve is observed at maturities ranging from 1 to 30 years, allowing for a detailed characterisation of the term structure of interest rates. High-grade private borrowing conditions are proxied using AAA-rated yield curve components at 5-, 10-, and 20-year maturities. All bond market variables are obtained from the European Central Bank Statistical Data Warehouse.

**Equity market data:** Equity market conditions are captured using both broad and sector-specific indicators. The dataset includes the Euro Stoxx 50 and the STOXX Europe 600 indices, expressed in levels and monthly percentage changes, which measure overall euro area and European equity market performance. To analyse heterogeneity across sectors, sectoral equity ETFs are included for key industries such as banking, utilities, technology, and oil and gas. These variables allow for a more granular assessment of equity market responses to monetary policy developments. Equity market data were sourced from Investing.com which aggregates data from major exchanges and index providers.

**Monetary policy measures:** Monetary policy conditions are measured using a set of conventional and unconventional indicators. The stance of conventional policy is captured by the ECB policy rate and the Overnight Index Swap (OIS) curve, observed at maturities from 1 month to 20 years, which reflects market expectations of future short-term interest rates. The size of the ECB balance sheet is measured by securities held for monetary policy purposes. In addition, the dataset includes a variable capturing the number of monetary policy-related events per month, reflecting the intensity of ECB policy actions and communication. All monetary policy variables are obtained from the European Central Bank.

**Control variables:** To account for macroeconomic and financial conditions, the dataset includes several control variables. Real economic activity is proxied by real GDP, measured in chain-linked volumes with 2010 as the reference year and observed at quarterly frequency. Inflation is measured using the Harmonised Index of Consumer Prices (HICP) for the euro area. Financial market uncertainty is captured by the VSTOXX index, which measures implied volatility in euro area equity markets and serves as a proxy for time-varying risk aversion. Macroeconomic data are sourced from Eurostat, while the VSTOXX index is provided by STOXX.

**Unconventional monetary policy variable:** The baseline shock variable (*qe_release_m*) corresponds to the QE factor from the Euro Area Monetary Policy Database (EA-MPD) of Altavilla et al. (2019), which captures high-frequency surprises in the ECB's balance sheet policy. Although labelled QE factor in the Altavilla et al. (2019) paper, it is used here as a broader unconventional monetary policy shock, encompassing both asset purchase expansions (QE) and contractions (QT) depending on the regime considered. It is extracted via rotated principal component analysis on high-frequency OIS yield changes in a narrow window around

ECB press releases, following the methodology of Swanson (2017). The factor loads primarily on long-term rates and captures unexpected changes in the ECB's asset purchase intentions as perceived by markets at the time of the announcement. It is expressed in basis points. By construction, the high-frequency identification strategy ensures that the variable captures only the unexpected component of ECB announcements within a ±10-minute window, making it exogenous to macroeconomic developments and thus free from endogeneity concerns (Altavilla et al., 2019).

## 3.2 Variables Construction

Since the economic interpretation of an unconventional monetary policy surprise depends on the prevailing policy stance, we interact *qe_release_m* with a set of mutually exclusive regime dummies that partition the sample period following the literature on ECB asset purchases. Table 1 summarises these periods and their sources. For each programme, the start date corresponds to the announcement date rather than the date of the first asset purchase, so as to fully capture the market surprise at the time of the policy decision. These dummies are the following:

- $D^{QE\_early} = 1$ during the early unconventional policy period (May 2009-October 2012; September-December 2014);
- $D^{QE} = 1$ during the full QE period (January 2015-December 2018; September 2019-February 2022);
- $D^{QT} = 1$ during the Quantitative Tightening period (June 2022 onward);
- $D^{None} = 1$ otherwise, i.e. periods with no active APP.

The final shock variables are constructed as the interaction of each dummy with *qe_release_m*:

$$QE\_shock_t \equiv D_t^{QE} \times qe\_release\_m_t \tag{1}$$
$$QE\_early\_shock_t \equiv D_t^{QE\_early} \times qe\_release\_m_t \tag{2}$$
$$QE\_Full\_shock_t \equiv \left(D_t^{QE\_early} + D_t^{QE}\right) \times qe\_release\_m_t \tag{3}$$
$$QT\_shock_t \equiv D_t^{QT} \times qe\_release\_m_t \tag{4}$$

This approach allows us to identify whether the market impact of a given asset purchase surprise differs across policy regimes, and in particular whether the transmission of unconventional monetary policy shocks is asymmetric between expansion and tightening phases.

## 3.3 Dependent Variables

We consider two families of dependent variables at monthly frequency. Given the local projection framework, both are measured as cumulative outcomes over horizons $h \in \{0, 1, 3, 6, 12\}$ months.

**Sovereign bond risk premia:** The primary bond-market outcome is the 10-year sovereign yield spread relative to Germany, which serves as the euro area risk-free benchmark:

$$\text{Spread}_t^c \equiv y_{10y,t}^c - y_{10y,t}^{DE}, \qquad c \in \{\text{IT, ES, PT, FR}\}, \tag{5}$$

where $y_{10y,t}^c$ denotes the 10-year government bond yield for country $c$. These spreads proxy for time-varying sovereign credit risk and euro area fragmentation risk. To capture market reactions to monetary policy shocks rather than persistent level differences, the dependent variable in the local projections is the cumulative change in the spread over horizon $h$, defined as $y_{t+h} - y_{t-1}$.

**Equity risk premia:** For equity markets, we measure realised equity risk premia as monthly excess returns on sectoral ETF indices:

$$\text{ERP}_t^s \equiv r_t^s - r_t^f, \tag{6}$$

where $r_t^s$ is the monthly return on sector $s \in \{\text{Technology, Utilities, Banks, Oil \& Gas}\}$ and $r_t^f = (1 + r_{ECB}^f/100)^{1/12} - 1$ is the monthly risk-free rate derived from the annualised ECB policy rate. Sectoral outcomes allow us to examine heterogeneity across industries that differ in cash-flow duration, a dimension emphasised by the portfolio-balance channel of unconventional monetary policy transmission. The dependent variable in the local projections is the cumulative equity risk premium over horizon $h$, defined as $\sum_{s=0}^{h} \text{ERP}_{t+s}^s$.

## 3.4 Descriptive Statistics

### 3.4.1 Interest rate dynamics and yield curve

**Long-term interest rates by country:** Figure 1 shows the 10-year government bond yields for France, Germany, Italy, Portugal, and Spain. A high degree of correlation exists between these countries, although significant divergence appears around 2012, particularly for Portugal due to the sovereign debt crisis and the loss of confidence from financial markets. After a long period of decline leading to near-zero or negative rates around 2020, yields show a sharp and synchronized increase starting in 2022, reflecting a global shift in monetary policy.

**Euro area yield curve spot rates:** Figure 2 displays the evolution of spot rates for maturities ranging from 2 to 30 years. The graph shows a typical upward-sloping yield curve for most of the period, where long-term rates remain higher than short-term rates. However, all maturities follow a similar downward trend until 2021 before rising sharply together. In recent years, the compression of the space between these lines indicates a flattening of the curve, and the 2-year rate even rises above all longer-term rates, signaling an inversion of the yield curve.

**Yield curve slope (10Y - 2Y spread):** Figure 3 highlights the slope of the yield curve, calculated as the difference between the 10-year and 2-year spot rates. The spread remains positive for the majority of the timeframe, peaking near 2.5 percentage points after 2008. A major shift occurs in late 2022, as the spread turns negative. This inversion of the yield curve is a significant macroeconomic signal, as it suggests that short-term interest rates are higher than long-term expectations, often serving as an indicator of an upcoming economic slowdown.

### 3.4.2 Equity market variation (STOXX Indices) and inflation

**Monthly variation of Stoxx 600 vs Stoxx 50:** Figure 4 illustrates the monthly percentage variations for the Stoxx 600 and Stoxx 50 indices. The two indices exhibit a very high correlation, as they move almost perfectly in sync throughout the entire period. Several periods of extreme volatility are visible, notably during the 2008 financial crisis and the 2020 COVID-19 pandemic shock, where monthly variations dropped below -10%. The graph shows that while the returns oscillate around the zero-percent line, the magnitude of fluctuations remains significant, reflecting the constant adjustment of equity markets to macroeconomic news and European monetary policy. The similarity between the two curves suggests that the systemic risk affecting the fifty largest European companies (Stoxx 50) is also representative of the broader market (Stoxx 600).

**HICP: Original index vs. detrended series:** Figure 5 illustrates the Harmonised Index of Consumer Prices (HICP) and its monthly variation. The original index follows a constant upward trend before a sharp acceleration in recent years. To ensure stationarity for the econometric analysis, the series is detrended using log-differences, which represent monthly inflation. The detrended graph shows that price instability increases significantly toward the end of the period, moving away from the stability observed in previous years.

### 3.4.3 Unconventional monetary policy shocks

**Descriptive statistics of unconventional monetary policy shocks:** Table 2 presents descriptive statistics for the four unconventional monetary policy shock variables over the full sample (September 2004 – September 2025, $T = 253$). All variables are expressed in basis points and are constructed as the interaction of *qe_release_m* with the corresponding regime dummy defined in equations (1)-(4). All shock variables have a mean and median close to zero, consistent with the rational expectations assumption underlying the high-frequency identification strategy. The median is exactly zero for all variables, which is mechanical: each shock is non-zero only in months where both an ECB meeting occurred and the corresponding regime dummy equals one, leaving the majority of observations at zero.The QT shock displays a substantially larger standard deviation (1.08 bp) and wider range (−9.58 to 8.71 bp) relative to the QE variables, suggesting greater volatility of balance sheet surprises during the tightening period. By contrast, the QE shocks show more contained dispersion, consistent with a more gradual and well-communicated asset purchase expansion.

#### 3.4.4 Sectoral performance (ETFs)

**Historical evolution of European sector ETFs:** Figure 6 tracks the price evolution of the selected sectors from 2005 to 2025. The Technology sector shows the most significant growth, characterized by a steady upward trend that accelerates after 2015. In contrast, the Banks sector experiences a major decline following the 2008 financial crisis and remains well below its initial levels for over a decade. The Utilities and Oil & Gas sectors show more moderate growth paths. These divergent trajectories suggest that sectoral reactions to macroeconomic shifts are highly heterogeneous in terms of magnitude while generally following a similar overall trend.

**Monthly percentage variation of sector ETFs:** Figure 7 displays the monthly price variations for each sector, providing a clearer view of short-term price instability. All sectors exhibit high levels of variation dispersion, with synchronized sharp drops during major global shocks, such as the 2008 crisis and the 2020 pandemic. The Technology and Oil & Gas sectors frequently show the largest monthly swings, while the Utilities sector generally presents more contained variations. This graph confirms that even when long-term trends differ, these sectors remain sensitive to common systemic shocks affecting the European financial market.

#### 3.4.5 Correlation analysis

Figure 8 presents the correlation matrix for the main variables used in the analysis. Several patterns are worth noting. The two QE shock variables are strongly correlated, as *QE (From 2015)* is a sub-period of *QE (From 2009)*. By construction, the QT shock is orthogonal to both QE variables, reflecting the mutual exclusivity of the regime dummies. Inflation and the policy rate are strongly positively correlated, consistent with the ECB tightening cycle, and the QT shock covaries with both, confirming that the tightening regime coincides with the high-inflation period in the sample. The sectoral equity risk premia are positively correlated with each other, suggesting common market drivers. The four sovereign spreads co-move strongly, reflecting the well-documented synchronisation of euro area peripheral sovereign risk. Sovereign spreads are negatively correlated with equity risk premia, consistent with a flight-to-quality dynamic during stress periods. Overall, the absence of extreme collinearity between the main explanatory variables supports the validity of the econometric estimations presented below.

# 4 Empirical Strategy

## 4.1 Econometric Framework: Panel Local Projections

To characterise the dynamic response of outcome variables to monetary policy shocks, we estimate Jordà-style local projections. This approach does not require the specification of the full multivariate dynamic data-generating process, is robust to misspecification, and directly yields horizon-specific estimates of cumulative effects without imposing the recursive structure of a VAR.

### 4.1.1 Panel Specification

To increase statistical power and exploit cross-sectional variation, we pool outcome series across countries (for sovereign spreads) and across sectors (for equity risk premia). The baseline panel local projection at horizon $h$ is:

$$y_{i,t+h} - y_{i,t-1} = \alpha_i^h + \beta_{QE}^h QE_t + \beta_{QT}^h QT_t + \theta^h MP_t^{short} + \Gamma^{h\prime} X_{t-1} + \varepsilon_{i,t}^h, \tag{7}$$

where $i$ indexes countries ($c \in \{IT, ES, PT, FR\}$) in the bond-spread panel or sectors ($s \in \{\text{Tech, Util, Bank, OilG}\}$) in the equity panel, and $h \in \{0, 1, 3, 6, 12\}$ denotes the forecast horizon in months. The dependent variable $y_{i,t+h} - y_{i,t-1}$ measures the cumulative change in the spread from the month preceding the shock to $h$ months after. For equity risk premia, the dependent variable is defined as the cumulative excess return from month $t$ to month $t+h$:

$$\sum_{j=0}^{h} ERP_{t+j}^{s}. \tag{8}$$

The entity fixed effects $\alpha_i^h$ absorb time-invariant heterogeneity in the average level and volatility of each series. In the sovereign-spread panel, these capture structural differences in sovereign risk across countries. In the equity panel, they absorb differences in average sectoral risk premia. The coefficients $\beta_{QE}^h$ and $\beta_{QT}^h$ are common across entities, measuring the average response of all countries or sectors to the respective monetary policy shock. The control vector $X_{t-1}$ includes lagged inflation (log-differenced Harmonised Index of Consumer Prices) and the ECB policy rate. We exclude the VSTOXX implied volatility index from the baseline specification due to extensive missing observations (115 out of 253 months), which would substantially reduce the effective sample size. Robustness checks including VSTOXX on the available subsample are discussed separately.

### 4.1.2 Inference

Standard errors are computed using the Newey and West (1987) heteroskedasticity- and autocorrelation-consistent (HAC) estimator with 6 lags. This correction accounts for conditional heteroskedasticity in monthly financial data and for the serial correlation mechanically induced by overlapping dependent variables at horizons $h > 0$.

### 4.1.3 Testing for Asymmetry

At each horizon $h$, the core hypothesis of symmetric effects is:

$$H_0: \ \beta^h_{QE} = \beta^h_{QT}. \tag{9}$$

We test this restriction using a Wald statistic. Let $\hat{\beta}$ denote the vector of estimated coefficients and $\hat{V}$ the Newey-West variance-covariance matrix. The restriction is expressed as $R\hat{\beta} = 0$ with $R = [0, \ldots, 1, -1, \ldots, 0]$, selecting the difference $\beta_{QE} - \beta_{QT}$. The two-sided $p$-value is computed as $2\Phi(-|z|)$, where $z = R\hat{\beta} / \sqrt{R\hat{V}R'}$. Rejection of $H_0$ at horizon $h$ indicates that QE and QT shocks of equal magnitude produce statistically different cumulative effects at that horizon. Comparing the individual significance levels of $\hat{\beta}^h_{QE}$ and $\hat{\beta}^h_{QT}$ is not a valid test of asymmetry: two coefficients may have similar point estimates yet differ in significance due to differences in estimation precision. The Wald test directly evaluates whether the difference $\beta^h_{QE} - \beta^h_{QT}$ is statistically distinguishable from zero, which is the relevant null hypothesis.

### 4.1.4 Entity-Level Projections

As a complement to the panel estimates, we report local projections estimated separately for each country and each sector:

$$y_{t+h} - y_{t-1} = \alpha^h + \beta^h_{QE} QE_t + \beta^h_{QT} QT_t + \theta^h MP^{short}_t + \Gamma^{h\prime} X_{t-1} + u^h_t. \tag{10}$$

These single-entity regressions allow us to examine cross-sectional heterogeneity in the response to monetary policy shocks. For instance, whether peripheral euro-area countries exhibit stronger sensitivity to QT shocks than core countries, or whether long-duration equity sectors respond differently from short-duration sectors. With approximately 208 observations per entity, these regressions have lower statistical power than the pooled panel and should be interpreted as exploratory.

## 4.2 Robustness: Conference-Window Shocks

As an additional robustness exercise, we re-estimate the panel local projections using monetary policy surprises measured during the ECB press-conference window rather than the press-release window. The conference-window shocks display substantially higher variance than the press-release shocks (standard deviation of 1.71 versus 0.43 for QE surprises) and are essentially uncorrelated with them (Pearson correlation of $-0.006$ for QE shocks). This suggests that most of the OIS variation during the press conference reflects noise from qualitative discussions rather than incremental policy signals about balance-sheet operations.

# 5 Main Results

## 5.1 Equity-Duration Heterogeneity

A central prediction of the portfolio-balance channel is that unconventional monetary policy should produce heterogeneous effects across assets that differ in cash-flow duration. Long-duration equities, whose valuations depend disproportionately on distant cash flows, are mechanically more sensitive to changes in long-term discount rates than short-duration equities. If QE compresses and QT decompresses term premia, one would therefore expect growth-oriented sectors to react more strongly than value sectors to balance-sheet policy surprises. To test this prediction, we augment the baseline panel local projection with an interaction between the monetary policy shocks and a continuous sector-level duration score $D_i \in \{0, 0.5, 1\}$:

$$\sum_{j=0}^{h} \mathrm{ERP}^{s}_{t+j} = \alpha_i^h + \beta^h_{QE} QE_t + \beta^h_{QT} QT_t + \delta^h_{QE}(QE_t \times D_i) + \delta^h_{QT}(QT_t \times D_i) + \theta^h MP_t^{short} + \Gamma^{h\prime} X_{t-1} + \varepsilon^h_{i,t} \tag{11}$$

We assign $D_i = 1$ to Technology (long-duration growth), $D_i = 0$ to Utilities (short-duration value), and $D_i = 0.5$ to Banks and Oil & Gas. The coefficients of interest are the interaction terms $\delta^h_{QE}$ and $\delta^h_{QT}$, which capture the marginal effect of equity duration on sensitivity to each regime's shocks.

The estimation yields no statistically significant duration heterogeneity at any horizon. The interaction coefficients $\hat{\delta}^h_{QE}$ and $\hat{\delta}^h_{QT}$ are economically negligible and statistically indistinguishable from zero across all horizons $h \in \{0, 1, 3, 6, 12\}$. At the impact horizon ($h = 0$), where duration effects should be most pronounced given the immediate repricing of future cash flows, the differential QT effect is $\hat{\delta}^0_{QT} = -0.0003$ (s.e. 0.0030). While point estimates at $h = 3$ are directionally consistent with the duration hypothesis (the cumulative QT effect increases monotonically from $-0.012$ for Utilities to $-0.016$ for Technology) the differential remains insignificant ($\hat{\delta}^3_{QT} = -0.004$, s.e. 0.003). This monotonic pattern breaks down at $h = 12$, where the QT effect on Technology loses significance while remaining significant for lower-duration sectors. Results are robust to restricting the panel to the two polar sectors (Technology and Utilities). The absence of significant cross-sectional heterogeneity suggests that QT shocks affect equity risk premia through a channel that operates uniformly across industries, rather than differentially compressing term premia along the cash-flow maturity structure. This finding is more consistent with the *risk-taking channel* of monetary policy: central bank balance-sheet operations appear to shift the aggregate price of risk rather than the relative pricing of long- versus short-duration assets. Under this interpretation, the withdrawal of central bank liquidity during QT triggers a broad reassessment of risk appetite that affects all equity sectors comparably, irrespective of their duration profile.

## 5.2 Asymmetric Effects on Sovereign Spreads

A key concern for the Eurosystem is the risk of financial fragmentation, whereby balance-sheet policy shocks disproportionately affect the borrowing costs of peripheral member states. The "scarcity" and "signalling" channels of QE predict that asset purchases compress sovereign spreads by providing an implicit backstop, while QT, by removing this marginal buyer, may produce a non-linear persistent decompression that is particularly pronounced for high-debt jurisdictions. We estimate the following panel local projection for the 10-year sovereign spreads of Italy, Spain, Portugal, and France relative to the German Bund:

$$\Delta^h \text{Spread}_{i,t+h} = \alpha_i^h + \beta_{QE}^h QE_t + \beta_{QT}^h QT_t + \theta^h MP_t^{short} + \Gamma^{h\prime} X_{t-1} + \varepsilon_{i,t}^h \tag{12}$$

The results reveal a striking regime-dependent asymmetry. At the impact horizon ($h = 0$), a one-basis-point hawkish surprise during a QE regime significantly compresses spreads ($\hat{\beta}_{QE}^0 = -0.0597$, s.e. 0.0339, significant at 10%), consistent with the view that unexpected tightening during an expansionary phase is interpreted as a signal of improving economic conditions, thereby reducing perceived fragmentation risk. In contrast, an identical shock during a QT regime produces a statistically negligible immediate effect ($\hat{\beta}_{QT}^0 = 0.0042$, s.e. 0.0030). The Wald test rejects symmetry at impact ($p = 0.058$).

This initial asymmetry deepens and reverses over the medium term. The compressive effect of QE shocks dissipates rapidly and becomes statistically insignificant from $h = 1$ onwards. Conversely, QT shocks generate an increasingly persistent spread widening: at $h = 3$, the cumulative effect reaches 0.029 pp (s.e. 0.009), significant at the 1% level, and remains significant at the one-year horizon ($\hat{\beta}_{QT}^{12} = 0.035$, s.e. 0.017). This divergent dynamic, transitory for QE, persistent for QT, is the central finding of this section and is consistent with a structural interpretation: while QE announcements produce temporary price adjustments that markets absorb within one to two quarters, QT surprises appear to induce a durable reassessment of sovereign credit risk that propagates well beyond the announcement quarter.

Individual country regressions confirm that this panel result is driven by peripheral sensitivity. Italy exhibits the strongest reaction to QT shocks, with a peak coefficient of 0.063 pp (s.e. 0.019) at $h = 3$. Spain and Portugal display intermediate and qualitatively similar responses. France, by contrast, remains almost entirely insulated from balance-sheet surprises across all horizons ($\hat{\beta}_{QT}^3 = 0.010$, s.e. 0.007), consistent with its status as a near-core sovereign. This cross-sectional heterogeneity underscores that the asymmetric effects of QT operate primarily through fragmentation risk rather than through a uniform repricing of euro-area sovereign debt. These results are illustrated in Figure 9.

## 5.3 Asymmetric Effects on Equity Risk Premia

We now turn to the equity market counterpart of the sovereign spread results. Section 3.1 established that the cross-sectional response of equity risk premia to monetary policy shocks is homogeneous across sectors; here, we focus on the temporal asymmetry in the persistence of QE versus QT shocks on the aggregate level of equity risk premia. The panel local projection for cumulative sectoral excess returns yields a clear and economically interpretable asymmetry. During the QT regime, hawkish surprises produce an immediate and highly persistent depression of equity risk premia: the cumulative effect on excess returns reaches $-0.0073$ (s.e. 0.0018) at $h = 1$ and $-0.0138$ (s.e. 0.0031) at $h = 3$, remaining statistically significant at the 5% level even at the twelve-month horizon ($\hat{\beta}^{12}_{QT} = -0.0130$, s.e. 0.0055). The impulse response function for QT shocks thus displays no meaningful mean-reversion over a one-year window.

The response under QE follows a strikingly different dynamic. Although a hawkish surprise during QE generates a significant short-run equity decline ($\hat{\beta}^{1}_{QE} = -0.0129$, s.e. 0.0068), the effect reverts quickly. By the twelve-month horizon, the QE coefficient turns positive ($\hat{\beta}^{12}_{QE} = +0.0124$, s.e. 0.0095) and is statistically indistinguishable from zero. The Wald test for long-run symmetry is strongly rejected at $h = 12$ ($p = 0.015$), confirming that QT shocks exert a far more durable depressive effect on equity valuations than equivalent QE shocks.

Sector-specific regressions identify the Banking and Technology sectors as the primary conduits of QT persistence. At $h = 3$, Banks display a sensitivity of $-0.023$ (s.e. 0.007) and Technology a sensitivity of $-0.019$ (s.e. 0.004) to QT shocks. These two sectors combine elevated sensitivity to financing conditions; Banks through net interest margins and credit risk; Technology through long-duration valuation multiples, which may explain their disproportionate response. Oil & Gas, by contrast, appears structurally decoupled from ECB balance-sheet surprises across all horizons and regimes, likely reflecting the dominance of global commodity price dynamics over euro-area monetary conditions in determining sectoral valuations.

Taken together, the equity results are consistent with a *risk-appetite interpretation* of QT: the withdrawal of central bank liquidity appears to trigger a persistent reassessment of risk-taking by equity investors, whereas hawkish surprises during QE regimes are treated as transitory price adjustments absorbed within a few quarters. This asymmetry in the persistence of monetary policy transmission has direct implications for the calibration of quantitative tightening. If contractionary balance-sheet surprises generate more durable financial tightening than equivalent expansionary ones, the effective tightening per unit of balance-sheet reduction may be larger than historical QE episodes would suggest, a finding that echoes, in the European equity market, the asymmetry identified for US Treasury yields by Swanson (2021) in the expectations component of the yield curve. The corresponding impulse response functions are displayed in Figure 10.

## 5.4 Sensitivity to the QE Shock Definition

The baseline specification uses `QE_Full_shock`, which pools early unconventional operations (CBPP1, SMP, CBPP2, ABSPP/CBPP3 from 2009 to 2014) with the formal Asset Purchase Programme launched in January 2015. However, these two periods differ substantially in institutional design and market context. The pre-2015 operations were targeted interventions conducted during the sovereign debt crisis, often aimed at specific market segments and accompanied by conditionality. The APP, by contrast, represents large-scale, broad-based asset purchases comparable in scope and design to the Federal Reserve's QE programmes studied by Lloyd and Ostry (2024). Pooling the two periods may dilute the signal of balance-sheet policy surprises by mixing structurally different interventions.

To assess the sensitivity of our results to this choice, we re-estimate the panel local projections using *QE_shock*, which restricts the QE regime to the formal APP period ($D^{QE} = 1$ during January 2015 to December 2018 and September 2019 to February 2022). The QT shock variable remains unchanged. This yields 52 non-zero QE shock observations instead of 98, but with a cleaner mapping to the large-scale purchase programme that is the natural counterpart to QT.

**Sovereign spreads:** The results, reported in Table 7, reveal a markedly different picture for the QE coefficient. Whereas the baseline specification produces insignificant QE effects beyond $h = 0$, the APP-only specification yields positive and significant QE coefficients at horizons $h = 1$ through $h = 6$. At $h = 3$, a one-basis-point hawkish surprise during the APP regime is associated with a cumulative spread widening of 0.091 pp (s.e. 0.035, significant at the 5% level), roughly three times the magnitude of the equivalent QT effect (0.031 pp). The Wald test rejects symmetry at $h = 3$ ($p = 0.075$). This suggests that, conditional on the formal QE regime, hawkish surprises generate a stronger short-run spread reaction than during QT, consistent with the interpretation that markets are particularly sensitive to signals of reduced ECB support during an Active Purchase Programme. The QT coefficients are essentially unchanged relative to the baseline, confirming that the QT estimates are robust to the definition of the QE counterfactual.

**Equity risk premia:** The APP-only specification produces a striking strengthening of QE significance on equity markets (Table 8). At $h = 0$, the QE coefficient reaches $-0.029$ (s.e. 0.008, significant at the 1% level), an effect roughly eight times larger in magnitude than the corresponding QT coefficient ($-0.003$). The Wald test rejects symmetry at the 1% level at both $h = 0$ ($p = 0.002$) and $h = 1$ ($p = 0.001$). This reversal of the asymmetry pattern is notable: whereas the baseline specification suggests that QT effects dominate in persistence, the APP-only specification reveals that QE effects dominate in magnitude at short horizons. By $h = 12$, however, the QE coefficient reverts to near zero (+0.006, insignificant) while the QT coefficient remains significant ($-0.012$, $p < 0.05$), reproducing the persistence asymmetry observed in the baseline.

**Interpretation:** These results suggest that the baseline QE coefficient is attenuated by the inclusion of pre-2015 early unconventional operations, which were institutionally distinct from the APP and appear to have generated weaker or noisier market reactions. When the QE shock is restricted to the formal APP period, two complementary asymmetries emerge. First, a magnitude asymmetry: QE surprises during the APP produce substantially larger immediate effects on both spreads and equity risk premia than QT surprises of equal size, consistent with the heightened market sensitivity to ECB purchase signals during an active program. Second, a persistence asymmetry: despite their larger initial impact, QE effects dissipate within two to three quarters, whereas QT effects persist to the one-year horizon. This dual pattern is consistent with a model in which QE announcements trigger a sharp but temporary portfolio-rebalancing response that is progressively arbitraged away, while QT surprises induce a more gradual but durable reassessment of risk conditions that markets are slower to absorb.

# 6 Limitations and Extensions

## 6.1 Limitations

Three limitations deserve acknowledgement. First, the QT regime begins in June 2022 and overlaps with a period of sharp conventional rate hikes, which may confound the balance-sheet channel with the interest rate channel. Although we control for conventional monetary policy surprises through $MP_t^{short}$ and the ECB policy rate, residual contamination cannot be ruled out. Second, the QT regime covers only approximately 40 monthly observations, which limits the statistical power of QT estimates relative to QE estimates and calls for caution in interpreting the long-horizon results. Third, the exclusion of VSTOXX from the baseline specification means that time-varying risk aversion is not directly controlled for; if QT episodes systematically coincide with elevated uncertainty, the estimated QT coefficients may overstate the causal effect of balance-sheet surprises on risk premia.

## 6.2 Possible Extensions

Three extensions would naturally complement this analysis. First, decomposing sovereign spreads into credit risk and redenomination risk components would clarify the channel through which QT widens peripheral spreads. Second, exploiting cross-sectional variation in countries' ECB debt holdings at the onset of QT would provide cleaner identification of the scarcity channel. Third, replacing the coarse discrete duration score $D_i$ with a continuous, time-varying measure based on dividend discount models would provide a more demanding test of the portfolio-balance channel, and may uncover heterogeneity that the current specification is too imprecise to detect.

# 7 Conclusion

This paper examines whether Quantitative Easing and Quantitative Tightening have asymmetric effects on bond and equity risk premia in the euro area. Combining the high-frequency shock identification of Altavilla et al. (2019) with panel local projections (Jordà, 2005) over a sample spanning September 2004 to September 2025, we interact the ECB's balance-sheet surprise with mutually exclusive regime dummies to isolate QE and QT shocks, and estimate their dynamic effects on sovereign spreads and sectoral equity risk premia using a Jordà-style panel framework with Newey-West standard errors.

Our first main finding concerns sovereign bond markets. QE surprises compress spreads at impact, consistent with a positive signal about economic conditions reducing perceived fragmentation risk, but this effect dissipates within one quarter. QT surprises, by contrast, generate a persistent and statistically significant spread widening that extends to the one-year horizon. This asymmetry is driven almost entirely by peripheral economies: Italy exhibits the strongest reaction with a peak coefficient of 0.063 pp at $h = 3$, while France remains largely insulated, consistent with its near-core status. These findings suggest that QT operates primarily through a fragmentation channel specific to the euro area, whereby the withdrawal of the ECB's implicit backstop induces a durable reassessment of sovereign credit risk in vulnerable economies.

Our second main finding concerns equity markets. Hawkish surprises during QT produce an immediate and persistent depression of cumulative excess returns that remains statistically significant at the twelve-month horizon, whereas the equivalent effect under QE fully reverts within a few quarters, a difference confirmed by a Wald test strongly rejecting long-run symmetry ($p = 0.015$). The Banking and Technology sectors are the primary conduits of this persistence, while Oil & Gas appears structurally decoupled from ECB balance-sheet surprises. Notably, we find no evidence that these effects differ across sectors of varying cash-flow duration, pointing to a risk-taking channel that uniformly reprices the aggregate price of risk rather than differentially compressing term premia along the maturity structure of cash flows.

Our third finding emerges from a sensitivity analysis restricting the QE shock to the formal Asset Purchase Programme period. This specification reveals a dual asymmetry absent from the baseline: APP surprises generate substantially larger immediate effects than QT surprises on both sovereign spreads and equity risk premia, roughly three to eight times larger in magnitude at short horizons, yet fully revert within two to three quarters. QT effects, by contrast, persist to the one-year horizon regardless of the QE counterfactual definition. This pattern is consistent with a model in which QE announcements trigger a sharp but temporary portfolio-rebalancing response that markets progressively arbitrage away, while QT surprises induce a more gradual but durable reassessment of risk conditions.

Taken together, these results suggest that QT is not simply the reverse of QE. The asymmetry in persistence (transitory under expansion, durable under contraction) implies that the effective financial tightening per unit of balance-sheet reduction may exceed what historical QE episodes would predict. For the Eurosystem, this carries a direct implication: gradualism and transparent forward guidance are especially important during the normalisation phase, given that contractionary balance-sheet surprises appear to generate disproportionately lasting pressures on both sovereign fragmentation risk and equity valuations.

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# Appendix

Table 1: ECB Asset Purchase Regimes

| Programme | Nature | Start | End |
|---|---|---|---|
| – | None | September 2004 | April 2009 |
| CBPP1 | Early QE | May 2009 | June 2010 |
| SMP | Early QE | May 2010 | September 2012 |
| CBPP2 | Early QE | October 2011 | October 2012 |
| – | None | November 2012 | August 2014 |
| ABSPP | Early QE | September 2014 | December 2014 |
| CBPP3 | Early QE | September 2014 | December 2014 |
| APP | QE | January 2015 | December 2018 |
| APP Pause | None | December 2018 | September 2019 |
| APP Relaunch | QE | September 2019 | February 2022 |
| PEPP | QE | March 2020 | February 2022 |
| – | None | March 2022 | May 2022 |
| APP Reinvestment End | QT | June 2022 | Ongoing |
| PEPP Reinvestment End | QT | December 2023 | Ongoing |

## Descriptive Statistics

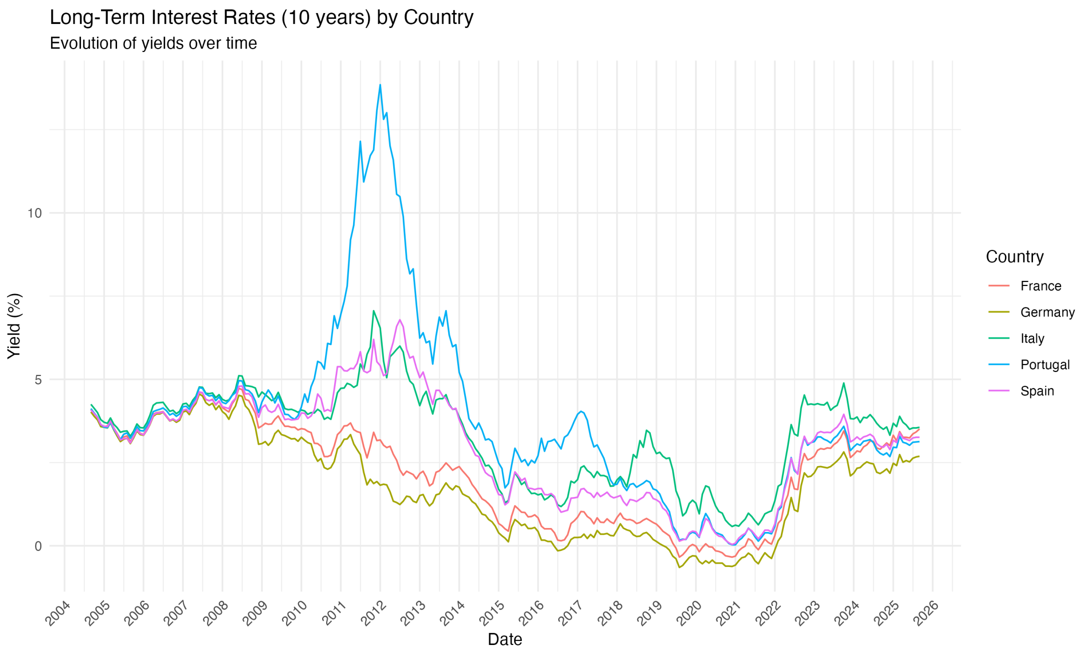


Figure 1: Long-Term Interest Rates (10 years) by Country

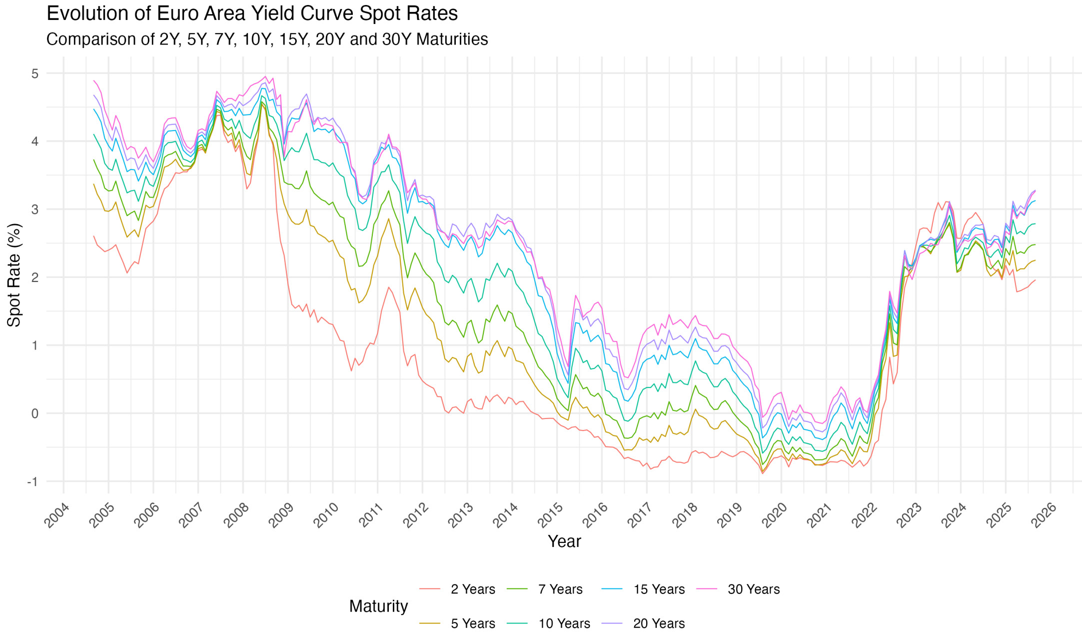


Figure 2: Evolution of Euro Area Yield Curve Spot Rates

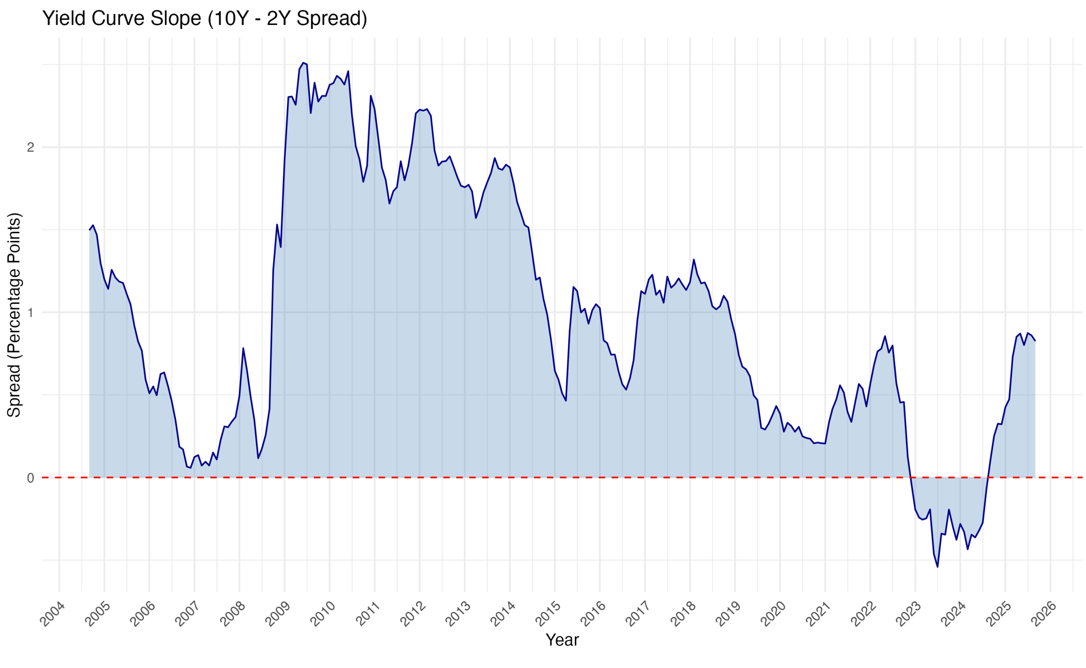


Figure 3: Yield Curve Slope (10Y - 2Y Spread)

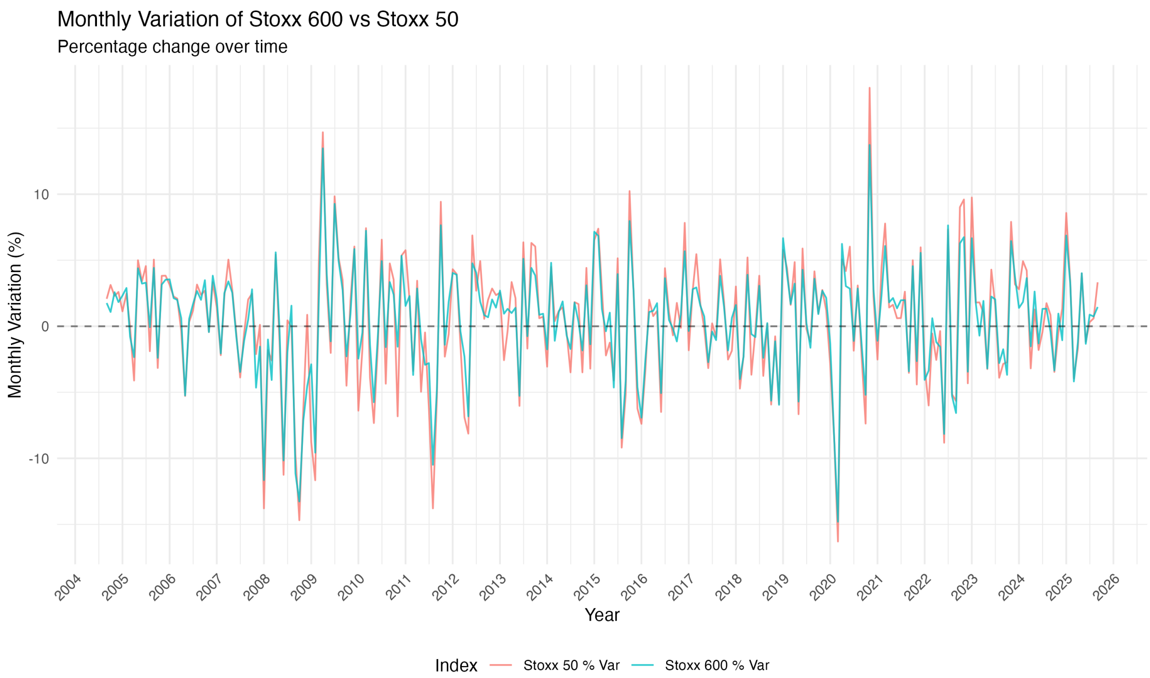


Figure 4: Monthly Variation of Stoxx 600 vs Stoxx 50

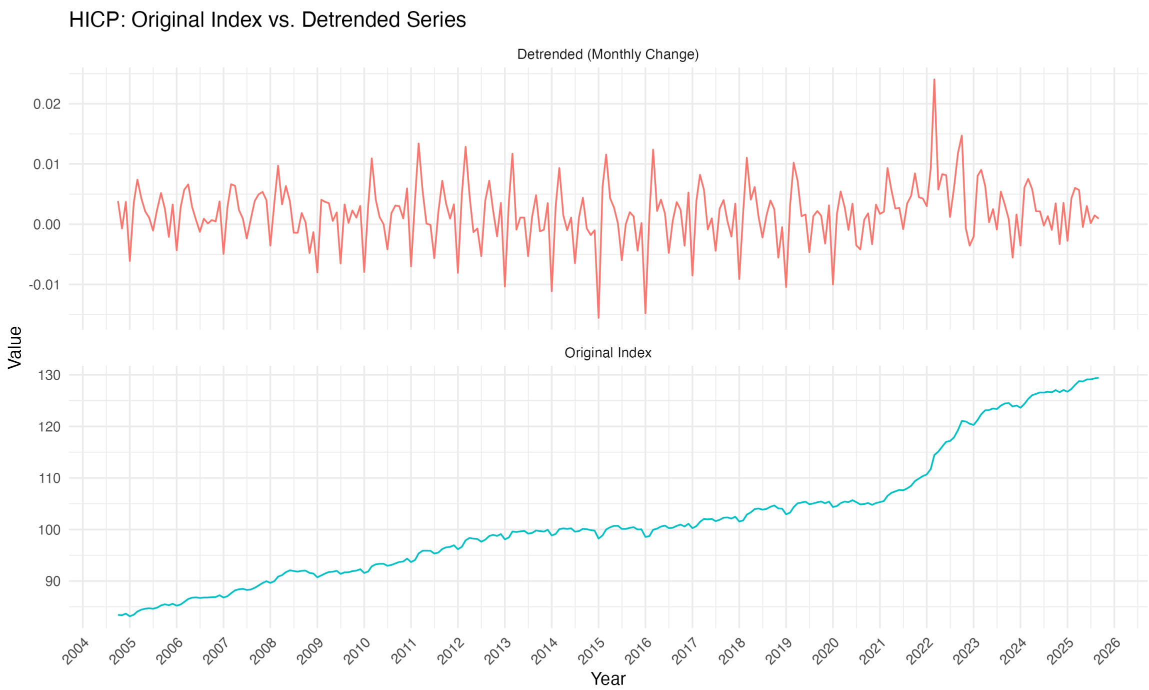


Figure 5: HICP: Original Index vs. Detrended Series

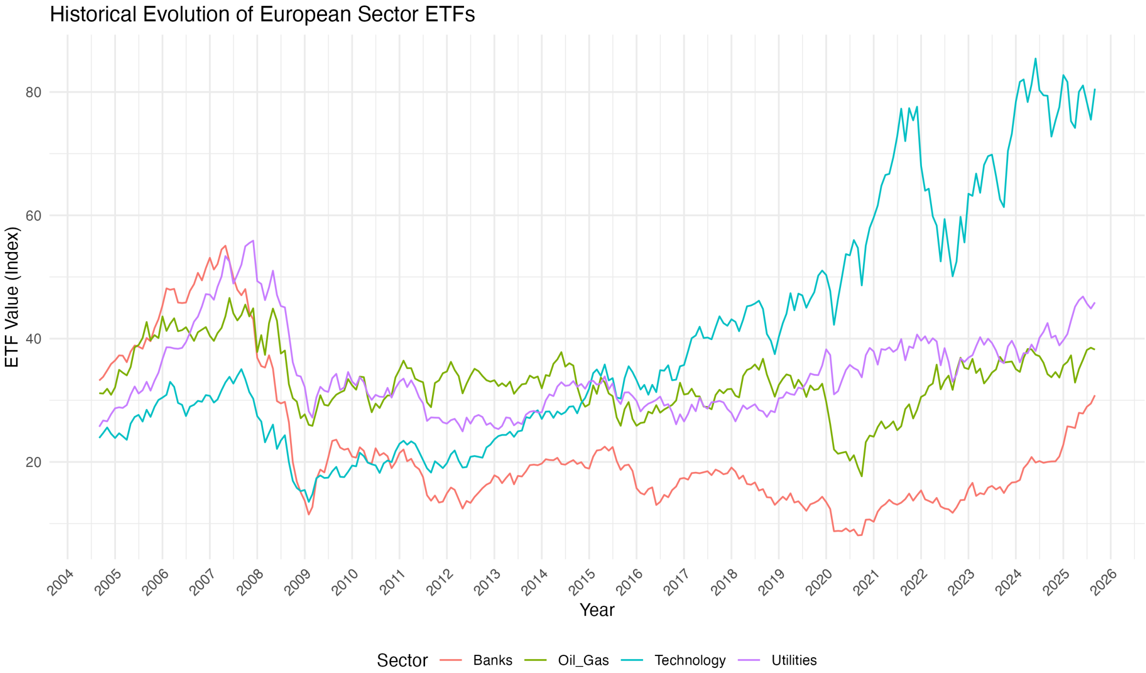


Figure 6: Historical Evolution of European Sector ETFs

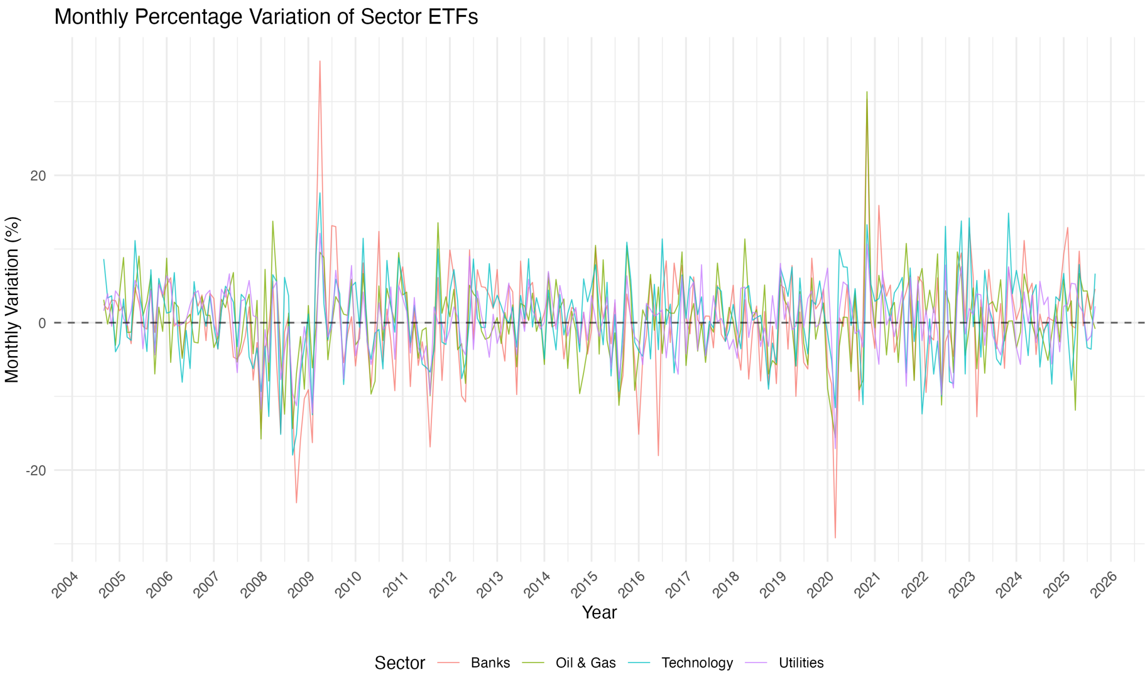


Figure 7: Monthly Percentage Variation of Sector ETFs

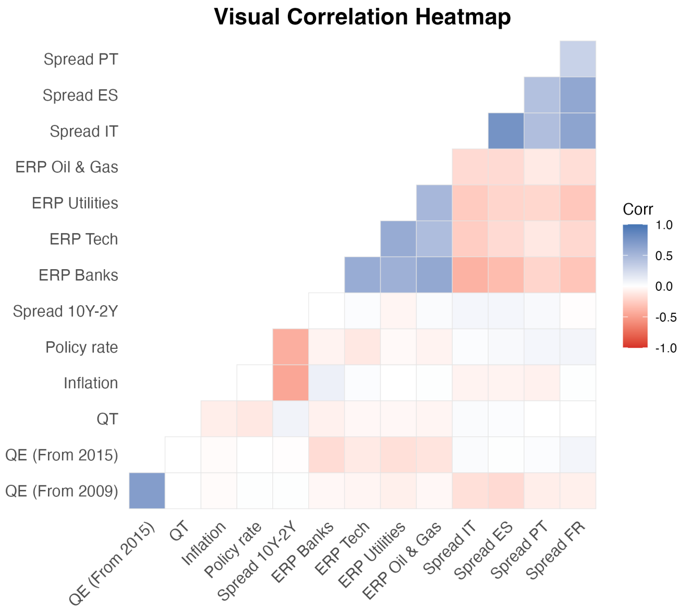


Figure 8: Correlation Matrix of Key Indicators

Table 2: Descriptive Statistics for Unconventional Policy Variable

| Variable | Obs | Mean | Std.Dev | Min | Median | Max |
|---|---|---|---|---|---|---|
| QE_Full_shock | 253.00 | -0.01 | 0.43 | -2.41 | 0.00 | 2.77 |
| QE_early_shock | 253.00 | -0.01 | 0.31 | -2.41 | 0.00 | 1.71 |
| QE_shock | 253.00 | 0.00 | 0.30 | -1.33 | 0.00 | 2.77 |
| QT_shock | 253.00 | -0.04 | 1.08 | -9.58 | -0.00 | 8.71 |

## Econometric Results

Table 3: Panel Local Projections — Sovereign Spreads (IT, ES, PT, FR)

| $h$ | $\hat{\beta}^h_{QE}$ | (s.e.) | $\hat{\beta}^h_{QT}$ | (s.e.) | Wald $p$-val | $N$ |
|---|---|---|---|---|---|---|
| 0 | $-0.0597^{*}$ | (0.0339) | 0.0042 | (0.0030) | $0.0584^{*}$ | 832 |
| 1 | $-0.0219$ | (0.0341) | $0.0167^{***}$ | (0.0049) | 0.2729 | 828 |
| 3 | 0.0053 | (0.0445) | $0.0291^{***}$ | (0.0091) | 0.6081 | 824 |
| 6 | 0.0453 | (0.0614) | $0.0263^{**}$ | (0.0109) | 0.7638 | 816 |
| 12 | 0.0923 | (0.0613) | $0.0349^{**}$ | (0.0168) | 0.3680 | 800 |

*Notes:* Panel LP at horizon $h$ (months) pooling Italy, Spain, Portugal and France. Dependent variable: cumulative change in 10-year sovereign spread relative to Germany (pp). Entity fixed effects included. Newey-West HAC standard errors (6 lags) in parentheses. Wald $p$-value tests $H_0 : \beta^h_{QE} = \beta^h_{QT}$. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 4: Panel Local Projections — Equity Risk Premia (Tech, Util, Bank, OilG)

| $h$ | $\hat{\beta}^h_{QE}$ | (s.e.) | $\hat{\beta}^h_{QT}$ | (s.e.) | Wald $p$-val | $N$ |
|---|---|---|---|---|---|---|
| 0 | $-0.0070$ | (0.0062) | $-0.0032^{**}$ | (0.0016) | 0.5506 | 832 |
| 1 | $-0.0129^{*}$ | (0.0068) | $-0.0073^{***}$ | (0.0018) | 0.4269 | 832 |
| 3 | $-0.0029$ | (0.0089) | $-0.0138^{***}$ | (0.0031) | 0.2475 | 832 |
| 6 | $-0.0120$ | (0.0117) | $-0.0107^{**}$ | (0.0044) | 0.9147 | 832 |
| 12 | $+0.0124$ | (0.0095) | $-0.0130^{**}$ | (0.0055) | $0.0149^{**}$ | 832 |

*Notes:* Panel LP at horizon $h$ (months) pooling Technology, Utilities, Banks and Oil & Gas. Dependent variable: cumulative sectoral equity risk premium from $t$ to $t+h$. Entity fixed effects included. Newey-West HAC standard errors (6 lags) in parentheses. Wald $p$-value tests $H_0 : \beta^h_{QE} = \beta^h_{QT}$. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 5: Individual Local Projections — Sovereign Spreads by Country

| $h$ | Country | $\hat{\beta}^h_{QE}$ | (s.e.) | $\hat{\beta}^h_{QT}$ | (s.e.) | $N$ |
|---|---|---|---|---|---|---|
| 4*0 | Italy | −0.0794 | (0.0873) | 0.0106 | (0.0077) | 208 |
| | Spain | −0.0754 | (0.0654) | 0.0050 | (0.0043) | 208 |
| | Portugal | −0.0698 | (0.0730) | 0.0008 | (0.0063) | 208 |
| | France | −0.0142 | (0.0268) | 0.0002 | (0.0030) | 208 |
| 4*1 | Italy | −0.0365 | (0.0769) | 0.0291*** | (0.0091) | 207 |
| | Spain | 0.0007 | (0.0458) | 0.0116 | (0.0073) | 207 |
| | Portugal | −0.0692 | (0.0997) | 0.0199 | (0.0135) | 207 |
| | France | 0.0175 | (0.0211) | 0.0060* | (0.0032) | 207 |
| 4*3 | Italy | −0.0184 | (0.0962) | 0.0626*** | (0.0190) | 206 |
| | Spain | 0.0194 | (0.0526) | 0.0279* | (0.0143) | 206 |
| | Portugal | 0.0109 | (0.1374) | 0.0166 | (0.0215) | 206 |
| | France | 0.0094 | (0.0282) | 0.0095 | (0.0065) | 206 |
| 4*6 | Italy | 0.0254 | (0.0999) | 0.0446** | (0.0213) | 204 |
| | Spain | 0.0422 | (0.0814) | 0.0212 | (0.0144) | 204 |
| | Portugal | 0.1086 | (0.2074) | 0.0321 | (0.0335) | 204 |
| | France | 0.0049 | (0.0358) | 0.0075 | (0.0076) | 204 |
| 4*12 | Italy | −0.0116 | (0.1079) | 0.0492 | (0.0396) | 200 |
| | Spain | −0.0188 | (0.1104) | 0.0524* | (0.0279) | 200 |
| | Portugal | 0.3865** | (0.1692) | 0.0284 | (0.0467) | 200 |
| | France | 0.0131 | (0.0258) | 0.0098 | (0.0070) | 200 |

*Notes:* Individual LP estimated separately for each country. Dependent variable: cumulative change in 10-year sovereign spread relative to Germany (pp). Newey-West HAC standard errors (6 lags) in parentheses. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6: Individual Local Projections — Equity Risk Premia by Sector

| $h$ | Sector | $\hat{\beta}^h_{QE}$ | (s.e.) | $\hat{\beta}^h_{QT}$ | (s.e.) | $N$ |
|---|---|---|---|---|---|---|
| 4*0 | Technology | −0.0042 | (0.0101) | −0.0017 | (0.0033) | 208 |
| | Utilities | −0.0092 | (0.0098) | −0.0022 | (0.0018) | 208 |
| | Banks | −0.0076 | (0.0174) | −0.0059 | (0.0039) | 208 |
| | Oil & Gas | −0.0071 | (0.0106) | −0.0029 | (0.0029) | 208 |
| 4*1 | Technology | −0.0084 | (0.0093) | −0.0087*** | (0.0033) | 208 |
| | Utilities | −0.0142 | (0.0101) | −0.0067*** | (0.0019) | 208 |
| | Banks | −0.0183 | (0.0188) | −0.0108** | (0.0048) | 208 |
| | Oil & Gas | −0.0108 | (0.0134) | −0.0029 | (0.0032) | 208 |
| 4*3 | Technology | 0.0056 | (0.0124) | −0.0185*** | (0.0040) | 208 |
| | Utilities | −0.0067 | (0.0099) | −0.0112*** | (0.0031) | 208 |
| | Banks | −0.0077 | (0.0274) | −0.0231*** | (0.0073) | 208 |
| | Oil & Gas | −0.0028 | (0.0174) | −0.0023 | (0.0049) | 208 |
| 4*6 | Technology | 0.0032 | (0.0217) | −0.0094 | (0.0072) | 208 |
| | Utilities | −0.0081 | (0.0108) | −0.0131** | (0.0058) | 208 |
| | Banks | −0.0308 | (0.0344) | −0.0169 | (0.0134) | 208 |
| | Oil & Gas | −0.0123 | (0.0195) | −0.0035 | (0.0057) | 208 |
| 4*12 | Technology | 0.0255 | (0.0212) | −0.0141 | (0.0109) | 208 |
| | Utilities | −0.0022 | (0.0141) | −0.0086 | (0.0069) | 208 |
| | Banks | 0.0124 | (0.0209) | −0.0217 | (0.0180) | 208 |
| | Oil & Gas | 0.0141 | (0.0196) | −0.0076 | (0.0059) | 208 |

*Notes:* Individual LP estimated separately for each sector. Dependent variable: cumulative equity risk premium from $t$ to $t+h$. Newey-West HAC standard errors (6 lags) in parentheses. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7: Panel Local Projections — Sovereign Spreads: APP-Only QE Shock

| $h$ | $\hat{\beta}^h_{QE}$ | (s.e.) | $\hat{\beta}^h_{QT}$ | (s.e.) | Wald $p$-val | $N$ |
|---|---|---|---|---|---|---|
| 0 | 0.0227 | (0.0305) | 0.0067** | (0.0032) | 0.5967 | 832 |
| 1 | 0.0701** | (0.0342) | 0.0188*** | (0.0053) | 0.1234 | 828 |
| 3 | 0.0911** | (0.0354) | 0.0306*** | (0.0095) | 0.0745* | 824 |
| 6 | 0.1012* | (0.0528) | 0.0266** | (0.0114) | 0.1403 | 816 |
| 12 | 0.0772 | (0.0651) | 0.0330* | (0.0170) | 0.4804 | 800 |

*Notes:* Panel LP at horizon $h$ (months) pooling Italy, Spain, Portugal and France. QE shock restricted to the formal APP regime ($D^{QE} = 1$: January 2015 to December 2018, September 2019 to February 2022). Dependent variable: cumulative change in 10-year sovereign spread relative to Germany (pp). Entity fixed effects included. Newey-West HAC standard errors (6 lags) in parentheses. Wald $p$-value tests $H_0 : \beta^h_{QE} = \beta^h_{QT}$. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8: Panel Local Projections — Equity Risk Premia: APP-Only QE Shock

| $h$ | $\hat{\beta}^h_{QE}$ | (s.e.) | $\hat{\beta}^h_{QT}$ | (s.e.) | Wald $p$-val | $N$ |
|---|---|---|---|---|---|---|
| 0 | −0.0292*** | (0.0084) | −0.0035** | (0.0016) | 0.0022*** | 832 |
| 1 | −0.0351*** | (0.0084) | −0.0074*** | (0.0018) | 0.0010*** | 828 |
| 3 | −0.0260** | (0.0120) | −0.0141*** | (0.0031) | 0.3305 | 824 |
| 6 | −0.0263 | (0.0198) | −0.0099** | (0.0041) | 0.4117 | 816 |
| 12 | +0.0062 | (0.0162) | −0.0121** | (0.0053) | 0.2721 | 800 |

*Notes:* Panel LP at horizon $h$ (months) pooling Technology, Utilities, Banks and Oil & Gas. QE shock restricted to the formal APP regime ($D^{QE} = 1$: January 2015 to December 2018, September 2019 to February 2022). Dependent variable: cumulative sectoral equity risk premium from $t$ to $t + h$. Entity fixed effects included. Newey-West HAC standard errors (6 lags) in parentheses. Wald $p$-value tests $H_0 : \beta^h_{QE} = \beta^h_{QT}$. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

## Impulse Response Functions

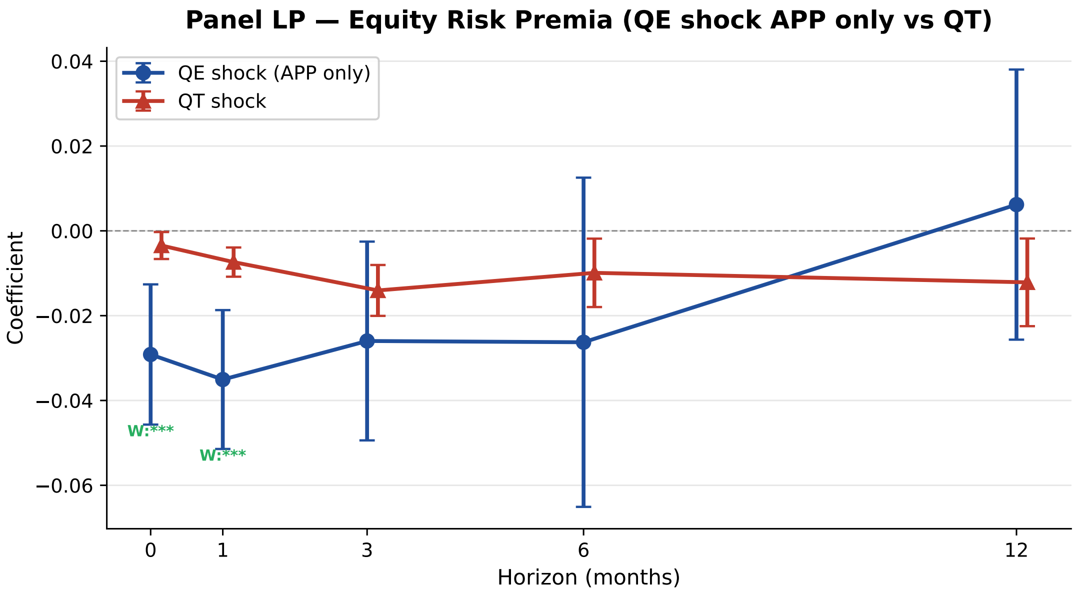


Figure 9: Impulse Response Functions from Panel Local Projections for 10-year Sovereign Spreads (Italy, Spain, Portugal, France relative to Germany)

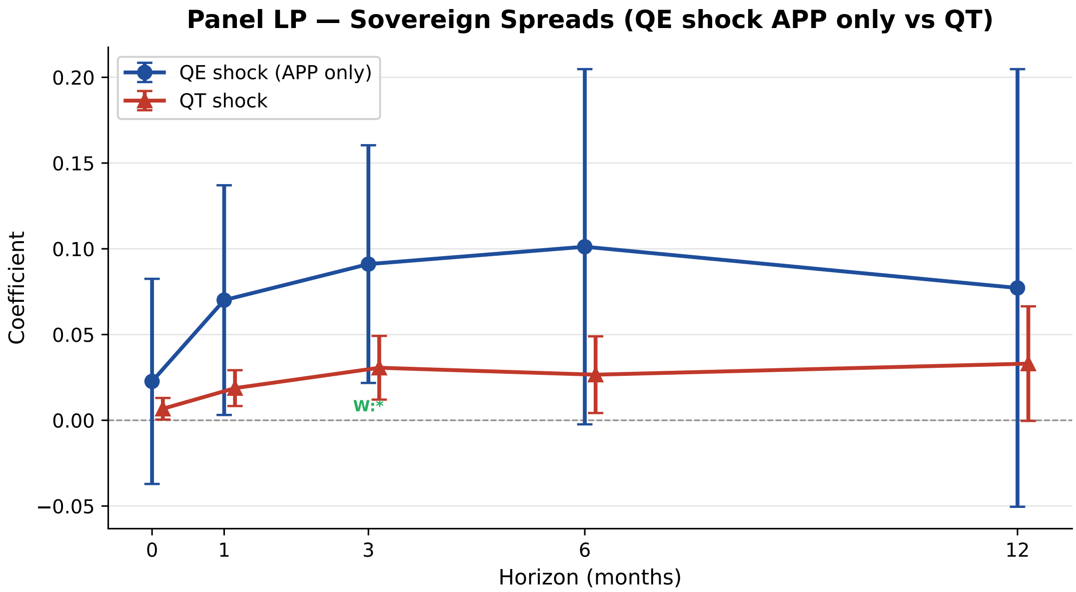


Figure 10: Impulse Response Functions from Panel Local Projections for Cumulative Sectoral Equity Risk Premia (Banks, Technology, Utilities, Oil & Gas)

## Codebook

| Variable | Type | Unit / Values | Description |
|---|---|---|---|
| ***A. Dates*** | | | |
| `Date` | Date | Month-Year | Observation date, monthly frequency. Sample: Sep 2004 - Sep 2025 ($T = 253$). |
| ***B. Sovereign Bond Markets*** | | | |
| `Longterm_IR_10Y_X` | Continuous | % (p.a.) | 10-year government bond yield for country $X \in$ {Spain, Deutschland, Italy, France, Portugal}. |
| `spread_X` | Continuous | pp | Sovereign spread of country $X \in$ {it, es, pt, fr} relative to Germany: $\text{spread}_X = r_{X,10Y} - r_{DE,10Y}$. |
| `d_spread_X` | Continuous | pp | First difference of `spread_X`: $\Delta\text{spread}_{X,t} = \text{spread}_{X,t} - \text{spread}_{X,t-1}$. |
| `Euro_Area_XY_Gov_Bond` | Continuous | % (p.a.) | Euro area government benchmark bond yield for maturity $XY \in$ {5Y, 10Y}. |
| ***C. Yield Curve*** | | | |
| `YC_spot_rate_NY` | Continuous | % (p.a.) | ECB zero-coupon spot rate at maturity $N \in$ {1, 2, 5, 7, 10, 15, 20, 30} years. |
| `yc_aaa_Ny` | Continuous | % (p.a.) | ECB par yield at maturity $N \in$ {5, 10, 20} years. AAA-rated euro area government bonds (changing composition). |
| `OIS_NM / NY` | Continuous | % (p.a.) | Overnight Index Swap (OIS) rate at maturity $N \in$ {SW, 1M, 3M, 6M, 1Y, 2Y, ..., 10Y, 15Y, 20Y}. SW = 1-week tenor. |
| ***D. Macroeconomic Controls*** | | | |

*(continued)*

| Variable | Type | Unit / Values | Description |
|---|---|---|---|
| `HICP` | Continuous | Index | Harmonised Index of Consumer Prices, euro area. |
| `policy_rate` | Continuous | % (p.a.) | ECB key policy interest rate. |
| `GDP` | Continuous | Index | Euro area real GDP index. |
| `VSTOXX` | Continuous | Index pts | Implied volatility index on the Euro Stoxx 50 (V2TX). Monthly value. |
| `Securities held…(ILM)` | Continuous | Millions EUR | ECB securities held for monetary policy purposes (Asset Purchase Programmes). |
| `n_events` | Discrete | 1, 2 | Number of ECB monetary policy meetings in the month. |
| ***E. Equity Markets*** | | | |
| `ETF_X_Value` | Continuous | Index pts | Monthly price level of the iShares STOXX Europe 600 ETF for sector $X \in$ {Banks, Utilities, Technology, OilGas}. |
| `Variation_in_%_ETF_X` | Continuous | % | Monthly percentage return of ETF for sector $X$. |
| `Stoxx600_Value` | Continuous | Index pts | Monthly price level of the STOXX Europe 600 index. |
| `Variation_in_%_Stoxx600` | Continuous | % | Monthly percentage return of the STOXX Europe 600. |
| `Stoxx50_Value` | Continuous | Index pts | Monthly price level of the Euro Stoxx 50 index. |
| `Variation_in_%_Stoxx50` | Continuous | % | Monthly percentage return of the Euro Stoxx 50. |
| ***F. Monetary Policy Shocks (EA-MPD)*** | | | |
| `target_m` | Continuous | bp | Target rate factor from ECB press releases, extracted via PCA on high-frequency OIS yield changes around monetary policy announcements (short-term rates). |

*(continued)*

| Variable | Type | Unit / Values | Description |
|---|---|---|---|
| fg_m | Continuous | bp | Forward guidance factor from ECB press releases, extracted via PCA on high-frequency OIS yield changes around monetary policy announcements (medium-term rates). |
| qe_release_m | Continuous | bp | QE factor from ECB press releases, extracted via PCA on high-frequency OIS yield changes around monetary policy announcements (long-term rates). |
| qe_conference_m | Continuous | bp | QE factor from ECB press conferences, extracted via PCA on high-frequency OIS yield changes around monetary policy announcements (long-term rates). |
| ***G. Regime Dummies*** | | | |
| D_None | Binary | 0/1 | = 1 during periods with no active APP: pre-May 2009, Nov 2012–Aug 2014, Jan–Aug 2019, Mar–May 2022. |
| D_QE_early | Binary | 0/1 | = 1 during early unconventional policy: May 2009–Oct 2012, Sep–Dec 2014. |
| D_QE | Binary | 0/1 | = 1 during full QE period: Jan 2015–Dec 2018, Sep 2019–Feb 2022. |
| D_QT | Binary | 0/1 | = 1 during Quantitative Tightening: Jun 2022 onward. |
| ***H. Interacted QE/QT Shocks*** | | | |
| QE_shock | Continuous | bp | $= D\_QE \times$ qe_release_m. QE surprise during full QE regime (press release). |

*(continued)*

| Variable | Type | Unit / Values | Description |
|---|---|---|---|
| QE_early_shock | Continuous | bp | $= D\_QE\_early \times$ qe_release_m. QE surprise during early QE regime. |
| QE_Full_shock | Continuous | bp | $= (D\_QE\_early + D\_QE) \times$ qe_release_m. QE surprise pooling early and full QE regimes. |
| QT_shock | Continuous | bp | $= D\_QT \times$ qe_release_m. QT surprise during tightening regime. |
| QE_shock_Conf | Continuous | bp | $= D\_QE \times$ qe_conference_m. Idem, from press conference. |
| QE_early_shock_Conf | Continuous | bp | $= D\_QE\_early \times$ qe_conference_m. |
| QE_Full_shock_Conf | Continuous | bp | $= (D\_QE\_early + D\_QE) \times$ qe_conference_m. |
| QT_shock_Conf | Continuous | bp | $= D\_QT \times$ qe_conference_m. |
| ***I. Constructed Risk Premia*** | | | |
| r_X | Continuous | Decimal | Monthly gross return of sector $X \in \{\text{tech, util, bank, oilg}\}$: Variation_in_percent /100. |
| rf_m | Continuous | Decimal | Monthly risk-free rate approximated from the ECB policy rate: $r^f_m = (1+r^f/100)^{1/12} - 1$. |
| erp_X | Continuous | Decimal | Equity risk premium for sector $X$: erp_X = r_X − rf_m. |

*Abbreviations:* bp = basis points (1 bp = 0.01%); pp = percentage points; p.a. = per annum (annualised rate); APP = Asset Purchase Programme; PCA = Principal Component Analysis; EA-MPD = Euro Area Monetary Policy Database (Altavilla et al., 2019); ECB = European Central Bank; OIS = Overnight Index Swap; ETF = Exchange-Traded Fund; AAA = triple-A credit rating.